where ρ_L = density of the liquid (which is not provided in the question!)

7 **A** Since the system is in equilibrium, the mass of the non-uniform rule can be found as 200 g – 64 g = 136 g.

Taking moments about 34.0 cm mark,
$136 \times d = 64 \times (34 - 4)$
$d = 14.1$ cm

Hence centre of mass = 34.0 + 14.1 = 48.1 cm.

8 **C** In Fig. 9, the vector sum of the tension force and the weight provides **the centripetal force** to